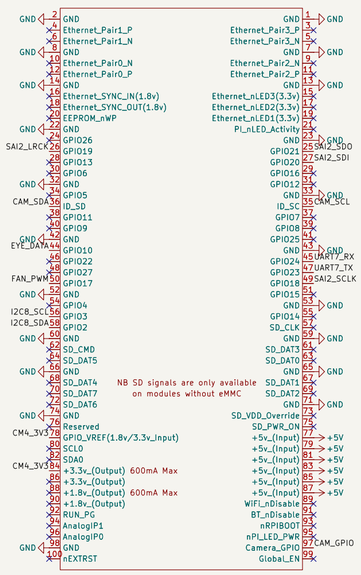
Radxa CM4 on DF40C sockets

Pin allocation: docs/head-pcb.md section 4. Unused pins not connected, as on the NANO-A

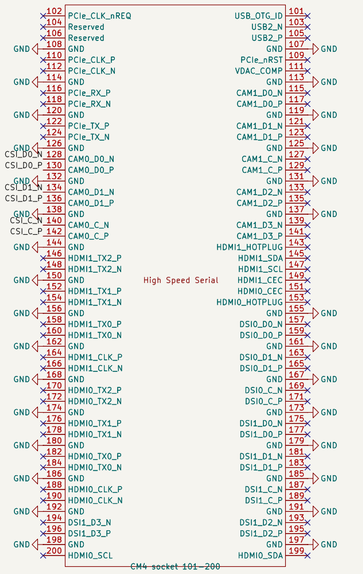
J9



NB SD signals are only available on modules without eMMC

CM4 socket 1–100

J10



High Speed Serial

CM4 socket 101–200

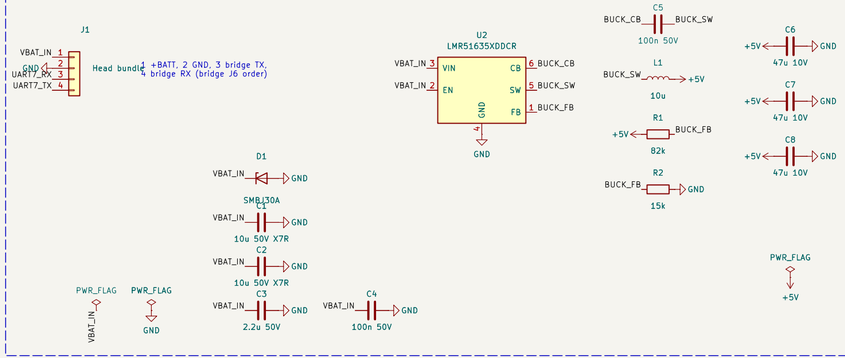
GPIO_VREF (78) on the module's own 3.3 V output (84; 86 is the same rail, left open): 3.3 V GPIO, as the NANO-A's H2 jumper

H5

Radxa CM4

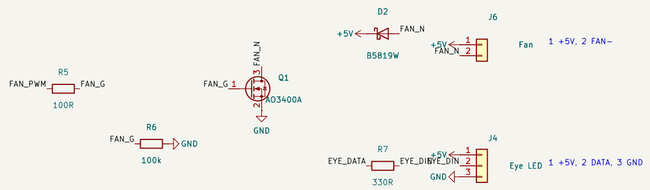
Head bundle in, 5 V buck

LMR51635X, 400 kHz, TI table 8-2 values. 82k/15k sets 5.17 V. EN tied to VIN. F6 (2 A) on the bridge fuses this feed



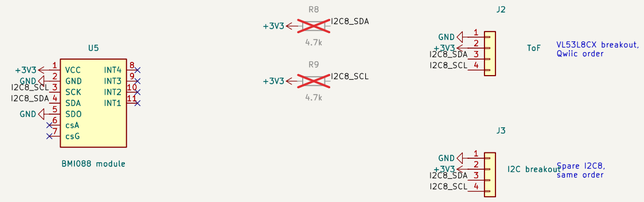
Fan and eye LED

Fan: low side, PWM1_CH2 (pin 50). Eye: WS2812-type, SPI1 MOSI (pin 44), 3.3 V data



I2C0: head IMU, ToF, breakout

Pull-ups on the modules (ToF 10k, BMI088 3.3k): R8, R9 not fitted. BMI088 SDO to GND: 0x18 and 0x68



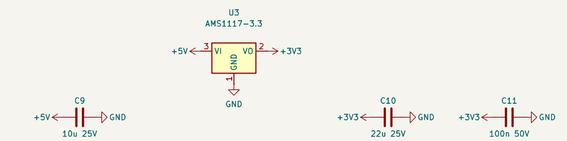
Power LED, mounting holes

M2 holes, one per corner, not connected



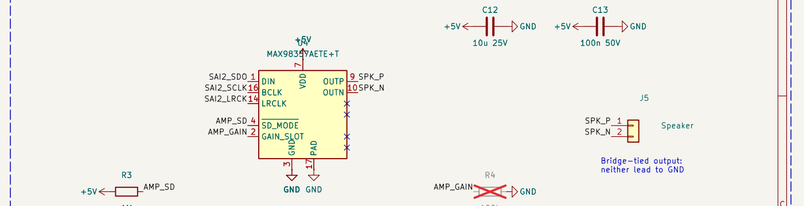
3.3 V

AMS1117-3.3 from 5 V, as on the NANO-A. Camera, ToF, IMU, mic, I2C breakout



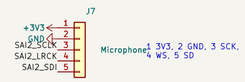
Speaker amplifier

MAX98357A on SAI2. SD_MODE 1M to 5 V: (L+R)/2 mono. GAIN open = 9 dB: fit R4 (100k to GND) for 15 dB



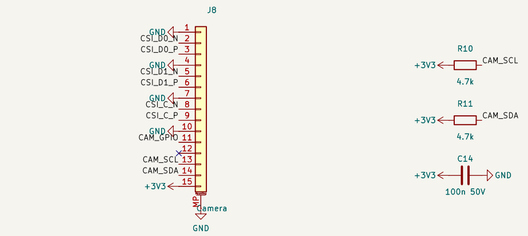
Microphone

INMP441 on SAI2. L/R tied to GND on the mic side (left slot)



Camera FPC

Pi 15-pin camera order. CAM0 lanes (Radxa CSI1), I2C6 with 4.7k pull-ups, CAM_GPIO enable. 100 ohm pairs



Bootstrapped by scripts/kicad/gen_head_sch.py; edit this file from now on
Design note: docs/head-pcb.md

Sheet: /
File: head.kicad_sch

Title: Dukki head carrier

Size: A2	Date: 12/10/2026	Rev: A, draft
KiCad E.D.A. 10.0.6		Id: 1/1